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COSC461 Project 1: Eagle Delivery
Due Oct 26th @ 11:59 PM

Eagle Delivery

Part A: Plan a Delivery with A* (30 Points)

(a) Formulate the Problem

The state is a tuple consisting of (row, column) and is where the robot currently is on the campus map. The initial state is a letter that corresponds to a specific building. The actions are up, down, left, or right. The transition model, or valid move is what changes the robot's position. Unless the robot attempts to make an action that would move it into an incorrect place, like into a wall, it will move into a pathway. The goal test is when the robot reaches its correct destination (another letter location). Each path cost is 1.

BFS is optimal for unit cost maps because the robot can move from shortest to longest path, and learn as it goes, instead of something like DFS which is expensive because it immediately raises the cost very quickly because the robot would be moving randomly towards an unknown goal.

(b) Why is the distance admissible in $h(s) = |r_s - r_g| + |c_s - c_g$?

Distance is admissible in the heuristic equation because it doesn't overestimate the cost and so the A* search remains in optimal conditions.

(c) Comparisons: Run BFS and A* for each of the six map and task combinations below:

Map	Delivery Task	BFS route len	A* route len	BFS exp nc	A* exp nc
Baseline	S -> W	18	18	67	22
Baseline	L -> P	16	16	57	18
Baseline	H -> D	22	22	66	32
Detour	S -> W	26	26	70	51
Detour	L -> P	16	16	51	18
Detour	H -> D	22	22	62	32

Explain whether the two algorithms find routes of equal length and how their expanded node counts compare. A* does not necessarily expand fewer nodes in every case.

While both BFS and A* both find routes of equal length because they're both designed to find

the most optimal path, BFS always has an expanded node count of more than A*, and actually nearly double A* in a majority of the cases. This is because A* has the advantage of using the Manhattan distance heuristic. Although BFS was more in every case, I believe it still has the possibility of being less than A*, it's just rarer.

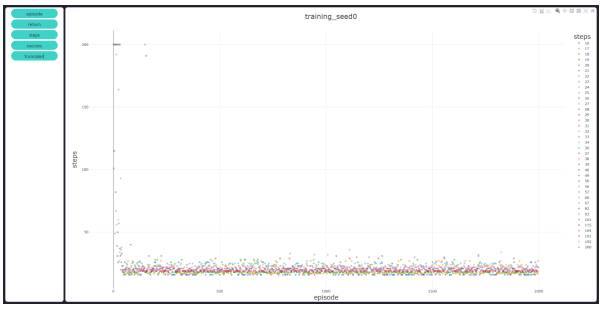
Part B: Learn a Delivery Policy (35 points)

(a) Examine the CSV files generated during training and evaluation for the required delivery tasks. Summarize your main observations using relevant measures, such as success rate and delivery steps.

Overall, the agent was able to get a 100% success rate across every delivery task. I wonder what the difference would be if we had a much larger map. During early training, step counts were higher than final because of greedy exploration but it leveled out well. All seeds overall had the same results.

Delivery Task	Success Rate	Mean Steps Success	Mean Return
S -> W	100%	18	22
L -> D	100%	22	18
H -> P	100%	16	24

(b) Create one or more plots that help you explain your results. Choose the measures and comparisons you find informative.



scatterplot_original

I used csvplot to visualize training_seed0 for H -> P. Other training_seed data would be similar so I didn't bother to do more than one, but it's a helpful way to visualize how quickly the step counts level out as more episodes occur. This is comparing steps (Y) to episodes (X)

Part C: Design Your Own Map (20 points)

(i) State your question and prediction before interpreting the results.

How does exploration rate affect training? I think the higher the rate (like 0.5), the more sporadic the agent will be and the less likely the plot levels out as quickly as a lower rate (like 0.2 or 0.05)

(ii) Explain the custom map: identify its loop, dead end, and relevant route choice.

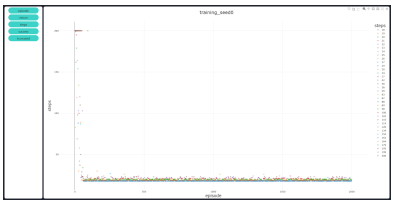
My custom map has a cyclic loop near the right side where the agent can potentially get trapped. It also has a dead end when going completely down and to the right. Overall the agent needs to quickly learn to avoid two major, expensive extremes.

(iii) Explain how you conducted your experiment, including what you changed, what you kept the same, and how you trained and evaluated the agent.

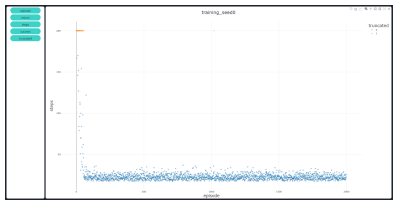
Overall I kept all the baseline variables the same except for exploration rate, where one was 0.05, one was 0.2, and the other was a much more extreme 0.5.

(iv) Summarize your findings using relevant results from the CSV files. Use tables, plots, or both to highlight the comparisons you find most informative. Clearly label the settings and measures being compared.

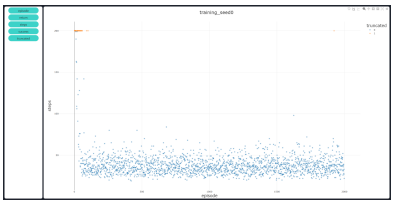
Overall, while maintaining the same overall shape and all reaching mean steps of 18 with a success of 22, my hypothesis was proven to be correct: The higher an exploration rate, the more scattered and sporadic the agent gets when exploring. 0.5 had a much smoother plot graph.



scatterplot_a



scatterplot_b



scatterplot_c

(v) Discuss what you learned and how the results compare with your expectations. Explain any interesting patterns or unexpected outcomes. Include unsuccessful runs or cases where changing a setting made little difference.

Overall, what I learned was that the results matched what I was expecting, but that was due to my prior knowledge from previous tests. Also, since AI agents and their search algorithms are meant to mimic human thinking, I'd figure it would probably be similar to how I think.

When exploration is low (like 0.05), training can sometimes be slow, but may be more optimal than high training (like 0.5) where the agent may go through cyclic loops or encounter dead ends more often.

Even when we do have extremes in exploration, typically time is the remaining factor that allows an agent's "knowledge" to eventually converge. Kind of poetic. If you just give yourself enough time, even if you're struggling, experience outpaces initial expectations.